



Assignment of bachelor's thesis

Title: Testing Neural Networks on Mobile Devices
Student: Hung Manh Nguyen
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Study program: Informatics
Branch / Artificial Intelligence 2021
specialization: Department of Applied Mathematics
Department: until the end of summer semester 2026/2027
Validity:

Instructions

The goal of this thesis is to experimentally evaluate the feasibility of running modern neural networks for face processing on mobile devices (Android, iOS). The student will compare different architectures, frameworks, and device types in terms of speed, energy consumption, and accuracy. The work will focus primarily on real-time face detection and facial landmark estimation from video.

Tasks

1. Familiarize yourself with the options for running neural networks on mobile platforms.
2. Select suitable models for facial landmark detection and collect several test videos.
3. Implement a testing scenario on both Android and iOS.
4. Evaluate latency (ms/frame), throughput (fps), and optionally energy consumption. Compare results across several different devices.
5. Optionally, explore various optimization methods (e.g., quantization) and evaluate their impact on speed and accuracy.

References

- [1] Li, Zhuojin, Marco Paolieri, and Leana Golubchik. "A benchmark for ML inference latency on mobile devices." Proceedings of the 7th International Workshop on Edge Systems, Analytics and Networking. 2024.
- [2] Lee, Juhyun, et al. "On-device neural net inference with mobile GPUs." arXiv

*Electronically approved by Ing. Magda Friedjungová, Ph.D.
on 13 February 2026 in Prague.*



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[3] Wang, Siqi, Anuj Pathania, and Tulika Mitra. "Neural network inference on mobile SoCs." IEEE Design & Test 37.5 (2020): 50-57.

[4] Suchanek, Matej. "Efficient Implementation of Neural Networks for Real-Time Applications." Bachelor's thesis. Czech Technical University in Prague, 2020.



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